

Birational Geometry: Exercise Collection

Exercise 1. Let X be a normal projective variety over $\mathbb{C}$ and let D be a Weil divisor.

1. Show that

$$H^0(X, D) := \{f \in K(X)^* \mid \operatorname{div}(f) + D \geq 0\} \cup \{0\}$$

is a $\mathbb{C}$ -subspace of $K(X)$.

2. Show that $\mathcal{O}_X(D)$ defined by

$$\mathcal{O}_X(D)(U) := H^0(U, D)$$

is a reflexive sheaf of rank 1.

3. Show that $D \mapsto \mathcal{O}_X(D)$ gives a 1-1 correspondence between

$$\{\text{Weil divisors}\} / \sim \longleftrightarrow \{\text{reflexive sheaves of rank 1}\} / \simeq.$$

Exercise 2. Let $f : X \rightarrow S$ be a proper surjective morphism between normal varieties with connected fibers, and let D, D' be Weil divisors. Show that the following two conditions are equivalent:

1. There exists a open covering $\{U_i\}_{i \in I}$ of S such that $D|_{X_i} \sim D'|_{X_i}$ for each i , where $X_i = f^{-1}(U_i)$.
2. There exists a Cartier divisor B on S such that $D \sim D' + f^*B$, or equivalently, exists an invertible sheaf $\mathcal{L}$ on S such that $\mathcal{O}_X(D) \simeq \mathcal{O}_X(D') \otimes_{\mathcal{O}_X} f^*\mathcal{L}$.

Exercise 3. Let $f : X \dashrightarrow Y$ be a rational map between normal proper varieties. Show that f is well-defined at any point of codimension 1 on X .

Exercise 4. Let $f : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$ be the (bi)rational map defined by $[x : y : z] \mapsto [1/x : 1/y : 1/z]$. Compute the exceptional locus of f and f^{-1} .

Exercise 5. Let $X = (xy = zw) \subset \mathbb{A}_{x,y,z,w}^4$. Let $\pi_1 : X_1 \rightarrow X$ be the blowup along $(x = z = 0)$, and let $\pi_2 : X_2 \rightarrow X$ be the blowup along $(x = y = z = w = 0)$. Compute the exceptional locus of π_1 and π_2 .

Exercise 6. Let $f : X \rightarrow Y$ be a projective birational morphism between normal varieties. Then clearly $f^{-1} : Y \dashrightarrow X$ is a birational map. Let P be a prime divisor on Y .

Show that f^{-1} is well-defined on the generic point of P ; moreover, there exists an open subset $V \subset Y$ such that $P \cap V \neq \emptyset$, $f^{-1} : V \rightarrow X$ is a morphism and $f^{-1} : V \rightarrow f^{-1}(V)$ is an isomorphism.

Exercise 7. Let $X = \mathbb{A}_{\mathbb{C}}^4$ and let $P \in X$ be the origin. Let $\pi : Y \rightarrow X$ be the blow-up at P and let E be the exceptional divisor. Consider X_1 to be the subvariety defined by $x_1x_2 + x_3x_4 = 0$ in X . Let $\pi_1 : Y_1 \rightarrow X_1$ be the induced blow-up and let E_1 be the exceptional divisor.

1. Compute the intersection number (E^4) on X .
2. Compute the intersection number (E_1^3) on X_1 .

Exercise 8. Let $f : X \rightarrow Y$ be a proper birational morphism between normal projective varieties and let D be an $\mathbb{R}$ -Cartier divisor on Y .

1. Show that D is big if and only if f^*D is big.
2. Let G, G' be $\mathbb{R}$ -Cartier divisors on X . Show that if $G \sim_{\mathbb{R}} G'$, then $f_*G \sim_{\mathbb{R}} f_*G'$ (pushforwards by birational **morphisms** preserve linear equivalence). Deduce from this that if G is big, then f_*G is big.

Exercise 9. Let Z be a proper variety of dimension k . Let $D_1, D_2, \dots, D_k$ be Cartier divisors on Z . Show that $\chi(Z, m_1D_1 + m_2D_2 + \dots + m_kD_k)$ is a polynomial in $m_1, \dots, m_k$.

If C is a curve in Z and $\nu : C^\nu \rightarrow C$ is the normalization. Show that

$$(D_1 \cdot C) = \deg_{C^\nu} \nu^*(\mathcal{O}_Z(D_1)|_C)$$

Exercise 10. Let X be a projective variety of dimension n and let A be an ample $\mathbb{R}$ -divisor on X . (Here an ample $\mathbb{R}$ -divisor means that we can write $A = \sum_{i=1}^k r_i A_i$ where $r_i > 0$ and A_i is a Cartier ample divisor for $1 \leq i \leq k$.)

Show that there exists a constant $c > 0$ and a positive integer m_0 such that

$$h^0(X, \lfloor mA \rfloor) > cm^n$$

for any $m \geq m_0$. (Hint: you should prove this statement first for a Cartier ample divisor A .)

Exercise 11. Let $f : Y \rightarrow X$ be a projective morphism between normal projective varieties. Let C be a curve on Y and let D be a Cartier divisor on X .

1. Use the Riemann–Roch formula to prove the projection formula:

$$(f^*D \cdot C) = (D \cdot f_*C).$$

2. Suppose that f is surjective and $\dim X = \dim Y = n$. Show that

$$\deg f \cdot (D^n) = ((f^*D)^n).$$

Exercise 12. Let X be a projective normal variety and let $A = \sum_{i=1}^k r_i A_i$ where $r_i \in \mathbb{R}_{>0}$ and A_i are Cartier ample divisors. Show that there exists a positive number $c > 0$ and a positive integer m_0 such that $h^0(X, mA) > cm^{\dim X}$ for all $m \geq m_0$. (It is not allowed to use the definition of big divisors.)

Exercise 13. Let $f : X \rightarrow Y$ be a projective birational morphism between normal projective varieties such that Y is $\mathbb{Q}$ -factorial.

1. Show that the exceptional set $\text{Exc}(f)$ is of pure codimension 1.
2. Suppose that X is $\mathbb{Q}$ -factorial. Denote $\text{Exc}(f) = \bigcup_{i=1}^k E_i$ where E_i are prime divisors. Show that

$$N^1(X)_{\mathbb{R}} = f^*N^1(Y)_{\mathbb{R}} \oplus \mathbb{R}[E_1] \oplus \mathbb{R}[E_2] \oplus \cdots \oplus \mathbb{R}[E_k].$$

Exercise 14. Let X be a projective smooth variety over $\mathbb{C}$. Suppose that $-K_X$ is ample. Show that $\chi(\mathcal{O}_X) = 1$. Furthermore, show that if $f : Y \rightarrow X$ is a finite étale morphism where Y is a projective variety, then f is an isomorphism.

Exercise 15. Let $\mathbb{P}^2$ be the projective plane and X be the blowup of $\mathbb{P}^2$ on two distinct point P_1, P_2 . Then there are 2 exceptional divisors E_1, E_2 on X corresponding to P_1, P_2 . On the other hand, take L to be the strict transform of the unique line passing P_1, P_2 .

1. Show that L, E_1, E_2 forms a basis of $N^1(X)_{\mathbb{R}}$.
2. Describe the nef cone and the pseudo-effective cone of X by L, E_1, E_2 .

Exercise 16. Prove the Bertini theorem for klt pairs: Let (X, B) be a klt pair and $|H|$ be a free linear system. Then for a general element $H \in |H|$, $(H, B|_H)$ is again klt.

Exercise 17. Let X be a smooth variety of dimension n and $x \in X$ be a closed point. Let B be an effective $\mathbb{R}$ -divisor on X . Show that

1. If $\text{mult}_x B > n$, then (X, B) is not lc.
2. If $\text{mult}_x B < 1$, then (X, B) is klt.

Here $\text{mult}_x B$ is defined as the following: for a prime divisor P , $\text{mult}_x P$ means the order of the local defining equation of P at x ; then linearly extend the definition to all $\mathbb{R}$ -divisors.

Exercise 18. Let $f : X \rightarrow Y$ be a projective birational morphism between normal projective varieties. Let D be an $\mathbb{R}$ -Cartier divisor on Y . Let F be an effective Weil divisor on X which is exceptional over Y .

Show that $\mathcal{O}_Y(D) \simeq f_*\mathcal{O}_X(\lfloor f^*D \rfloor + F)$.

Exercise 19. Let $f : X \rightarrow Y$ be a finite morphism between normal projective varieties. Suppose that B_X and B_Y are effective $\mathbb{R}$ -divisors on X, Y such that $f^*(K_Y + B_Y) = K_X + B_X$.

Show that (Y, B_Y) is KLT/PLT/LC iff (X, B_X) is KLT/PLT/LC.

Exercise 20. Let $h : X \rightarrow Y$ and $h' : X \rightarrow Y'$ be projective morphisms with connected fibers between normal varieties. Assume that a curve C on X is contracted by h (i.e., $h(C)$ is a point) if and only if C is contracted by h' .

Show that there exists an isomorphism $\tau : Y \rightarrow Y'$ commuting with h, h' .

Exercise 21. Let X be a normal projective variety over $\mathbb{C}$. Let D be a semi-ample Cartier divisor on X (i.e., mD is free for some positive integer m). Show that the section ring

$$\bigoplus_{k \geq 0} H^0(X, kD)$$

is finitely generated as a graded $\mathbb{C}$ -algebra.

Exercise 22. Show the following version of connectedness lemma: Let (X, B) be a pair where X is projective. Define $U \subset X$ to be the largest open subset such that $(U, B|_U)$ is KLT. Then $X \setminus U$ is called the non-KLT locus of (X, B) . Suppose that $-(K_X + B)$ is nef and big, show that the non-KLT locus of (X, B) is connected.

Exercise 23. Let (X, B) be a DLT pair and let Z_1, Z_2 be prime divisors such that $B \geq Z_1 + Z_2$. Suppose that $Z_1 \cap Z_2 \neq \emptyset$. Show that $Z_1 \cap Z_2$ has pure codimension 2 in X .

Exercise 24. Let (X, B) be a projective klt pair over $\mathbb{C}$. Suppose that $-(K_X + B)$ is nef and D is a big $\mathbb{Q}$ -Cartier divisor on X such that $D \cdot C > 0$ for any curve C on X . Show that D is ample.

Exercise 25. Let X be the blowup of $\mathbb{P}^2$ at 1 point. Describe the extremal rays of $\overline{\text{NE}}(X)$ and the contraction morphisms of those extremal rays.

Exercise 26. Let X be the blowup of $\mathbb{P}^2$ at 2 points. Describe the extremal rays of $\overline{\text{NE}}(X)$ and the contraction morphisms of those extremal rays.

Exercise 27. Let (X, B) be a $\mathbb{Q}$ -factorial DLT pair where X is projective over S . Suppose that a $(K_X + B)$ -MMP over S terminates. Show that it terminates at a minimal model if and only if $K_X + B$ is pseudo-effective over S , and it terminates at a Mori fiber space if and only if $K_X + B$ is not pseudo-effective over S .

Exercise 28. Let (X, B) be a $\mathbb{Q}$ -factorial DLT pair where X is projective over S . Let $(X, B) \dashrightarrow (X', B')$ be a step of MMP (a divisorial contraction or a flip). Show that (X', B') is also DLT.

Exercise 29. Let X be an affine variety.

1. Let $X \subset \bar{X}$ be any open embedding into a projective variety. Show that $\bar{X} \setminus X$ is of pure codimension 1.
2. Show that there exists $X \subset \bar{X}$ an open embedding into a projective variety, such that $\bar{X} \setminus X$ is the support of an ample divisor.

Exercise 30. Suppose that $X \rightarrow Y$ is a small extremal contraction and $X^+ \rightarrow Y$ is a flip of $X \rightarrow Y$. Show that $\rho(X^+/Y) = 1$.

Exercise 31. Let (X, B) be a $\mathbb{Q}$ -factorial DLT pair of dimension d where X is projective over S . Suppose that S is projective and H is a Cartier ample divisor on S . Show that $K_X + B$ is nef over S if and only if $K_X + B + 2df^*H$ is nef.

Exercise 32. Let (X, B) be a $\mathbb{Q}$ -factorial DLT pair where X is projective over S . Let $\phi : X \dashrightarrow X'$ be a flip. Let D be an $\mathbb{R}$ -Cartier divisor such that D and ϕ_*D are both nef over S . Show that f_*D is an $\mathbb{R}$ -Cartier divisor on Y and $D = f^*f_*D$ where $f : X \rightarrow Y$ is the contraction morphism of the flip.

Exercise 33. Let $A, B \subset V$ be closed convex cones in a finite-dimensional $\mathbb{R}$ -linear space. The dual of A is defined to be

$$A^* = \{u \in V^* \mid (u, v) \geq 0 \text{ for any } v \in A\}.$$

Show that

$$A^* + B^* = (A \cap B)^*$$

Exercise 34. Let X be a $\mathbb{Q}$ -factorial variety and $X \rightarrow S$ be a projective morphism. Suppose that (X, B) is DLT and $(X, B + D)$ is LC. Suppose that $K_X + B + D$ is nef and there exists a $(K_X + B)$ -negative extremal ray R such that $(K_X + B + D) \cdot R = 0$ and R induces a step of the MMP $\phi : X \dashrightarrow X'$ (X' is either the image of the divisorial contraction or the flip.) Then consider the pair $(X', B' + D' = \phi_*(B + D))$.

Show that $(X', B' + D')$ is again LC and $K_{X'} + B' + D'$ is again nef.

Exercise 35. Let X be a smooth projective variety and D be a numerically movable $\mathbb{R}$ -divisor such that $D \neq 0$. Show that $D \cdot H_1 \cdot H_2 \dots H_{n-1} > 0$ for any ample divisors as outlined below:

1. Show that $D \cdot H_1 \cdot H_2 \dots H_{n-1} \geq 0$;
2. Show that if an $\mathbb{R}$ -divisor E on X satisfies $E \cdot H^{n-1} = E^2 \cdot H^{n-2} = 0$ for some ample divisor H on X , then $E \equiv 0$. (Hint: use the Hodge index theorem on surfaces);
3. Show that $D \cdot H_1 \cdot H_2 \dots H_{n-1} \neq 0$ by (2).

Exercise 36. Let $(X, B) \rightarrow S$ be a $\mathbb{Q}$ -factorial DLT pair projective over S , and let $X \rightarrow Y$ be a small contraction over S with respect to a $(K_X + B)$ -negative extremal ray. Let B' be another $\mathbb{R}$ -divisor such that $-(K_X + B')$ is ample over Y . Show that a flip of (X, B') over Y is the same as a flip of (X, B) over Y .

Exercise 37. Show that (X, B) is DLT if and only if the following conditions hold:

1. the coefficients of B are at most 1;
2. there exists a closed subset $Z \subset X$ such that $(X \setminus Z, B|_{X \setminus Z})$ has log smooth support;
3. for any resolution $g : Y \rightarrow X$, and any prime divisor E on Y such that $g(E) \subset Z$, if we write $g^*(K_X + B) = K_Y + C$, then the coefficient of E in C is strictly smaller than 1.

Exercise 38. Let X be a $\mathbb{Q}$ -factorial variety and $X \rightarrow S$ be a projective morphism. Suppose that (X, B) and (X, B') are DLT pairs.

Suppose that (Y, B_Y) is a minimal model of (X, B) over S and $Y \rightarrow Z$ is the canonical model. Take B'_Y to be the strict transform of B' on Y . Suppose that $(Y', B'_{Y'})$ is a minimal model of (Y, B'_Y) over Z and $Y' \rightarrow Z'$ is the canonical model. Take $B_{Y'}$ to be the strict transform of B on Y' .

Show that there exists a real number $\delta > 0$ such that for any $0 < t < \delta$, $(Y', (1-t)B_{Y'} + tB'_{Y'})$ is a minimal model of $(X, (1-t)B + tB')$ over S and $Y' \rightarrow Z'$ is the canonical model.

Exercise 39. Let (X, B) be a DLT pair such that $[B] = Y$ is irreducible. Show that (X, B) is PLT and (Y, B_Y) is KLT, where (Y, B_Y) is given by the subadjunction formula $(K_X + B)|_Y = K_Y + B_Y$.

Exercise 40. Explain the detail of the argument that “existence of minimal models” implies “extension theorem”:

Let $(X, B) \rightarrow S$ be a pair satisfying the BCHM condition such that X is smooth and B has normal crossing support. Suppose that $[B] = Y$ is irreducible and S is affine. Suppose that (Y, B_Y) is terminal. Fix a positive integer $m_0 \geq 2$ such that $m_0(K_X + B)$ is a Weil divisor. Suppose that a general element of $|m_0(K_X + B)|$ (assumed to be non-empty) does not contain Y .

1. Show that Y is not contracted on a minimal model of $(X, B) \rightarrow S$;
2. Show that the natural map

$$H^0(X, mm_0(K_X + B)) \rightarrow H^0(Y, mm_0(K_Y + (B - Y)|_Y))$$

is surjective for any $m \geq 1$.

Exercise 41. Let $(X, B) \rightarrow S$ be a $\mathbb{Q}$ -factorial DLT pair projective over S , and let H be a $\mathbb{Q}$ -Cartier divisor ample over S such that $(X, B + H)$ is lc. Suppose that $H \equiv_S 2dH'$ where H' is Cartier and $d = \dim X$.

1. If H is ample over S , show that $K_X + B + H$ is nef over S .
2. If H is nef over S , show that every $(K_X + B + H)$ -MMP over S is a $(K_X + B)$ -MMP over S .

Exercise 42. Let X be a $\mathbb{Q}$ -factorial normal projective variety. Show that the Zariski–Nakayama decomposition of a pseudo-effective $\mathbb{R}$ -divisor D on X is well-defined.

Exercise 43. Let X be the blow-up of 2 points in $\mathbb{P}^2$. Describe the Zariski–Nakayama decomposition of all pseudo-effective $\mathbb{R}$ -divisors $D \in N^1(X)_{\mathbb{R}}$.

Exercise 44. Let (X, B) be a quasi-projective KLT pair. Assuming the existence of flips and termination of MMP with scaling under BCHM condition. Show that (X, B) has a $\mathbb{Q}$ -factorialization.

Exercise 45. Let $(X, B) \rightarrow S$ be a pair satisfying the BCHM condition. Suppose that it has a minimal model. Show that it has a canonical model. Show that $K_X + B \sim_{\mathbb{R}, S} D \geq 0$. (This is the implication that “existence of minimal model” implies “nonvanishing”, so the results of BCHM is not allowed to use)

Exercise 46. Show that the finite generation fails for some (1) $K_X + B$, where (X, B) is klt and B is an $\mathbb{R}$ -divisor; (2) $\mathbb{Q}$ -divisor D .

Exercise 47. Let $X \rightarrow Z$ be a flipping contraction where Z is affine. Show that for any irreducible divisor $S \subset X$ we have $S \sim S'$, where S' is an effective divisor whose support does not contain S .

Exercise 48. Show that if the flip exists for any klt pair (X, B) with $B \in \mathbb{Q}$, then the flip exists for any klt pair (X, B) with $B \in \mathbb{R}$.

Exercise 49. Let (X, B) be a projective KLT pair. Let $f : X' \rightarrow X$ be a log resolution and let B' be an effective divisor on X' such that (X', B') is KLT and

$$K'_{X'} + B' \equiv f^*(K_X + B) + E$$

where E is an effective f -exceptional $\mathbb{R}$ -divisor whose support contains all f -exceptional divisors.

Show that a minimal model of (X, B) is the same as a minimal model of (X', B') .

Exercise 50. Let X be a $\mathbb{Q}$ -factorial normal quasi-projective variety and let $f : X' \rightarrow X$ be a resolution. Let E be an effective f -exceptional $\mathbb{R}$ -divisor and let A be a divisor on X' ample over X . For a sufficiently small $0 \leq t \ll 1$, consider the Nakayama–Zariski decomposition $E + tA = P_t + N_t$. Show that $\text{Supp}(N_t) = \text{Supp}(E)$ and $N_0 = E$.

Exercise 51. Let (X, B) be a plt pair such that $Y = \lfloor B \rfloor$ is irreducible. Prove: there exists a log resolution $\mu : X' \rightarrow X$ of (X, B) , such that $B'^+ - Y'$ is a disjoint union of irreducible components, where $K_{X'} + B'^+ - B'^- = \mu^*(K_X + B)$, and $B'^+ \geq 0$ and $B'^- \geq 0$ have no common components.

Exercise 52. Let D be a Cartier divisor on a quasi-projective variety X , and Y a prime divisor on X . Suppose that $H^0(X, D) \rightarrow H^0(Y, D|_Y)$ is surjective, and $\text{Mov}(D)$ is free. Show that $\text{Mov}(D)|_Y = \text{Mov}(D|_Y)$.

Exercise 53. Let $X \dashrightarrow Y$ be a birational map surjective in codimension 1, where X, Y are normal $\mathbb{Q}$ -factorial varieties over S . Take a common log resolution $p : W \rightarrow X, q : W \rightarrow Y$. Suppose that D is an $\mathbb{R}$ -divisor such that $\alpha_* D$ is nef. Show that $p^*(D) - q^*(\alpha_* D) = G$ is effective if and only if $p_* G \geq 0$.

Exercise 54. Let $X \dashrightarrow Y$ be a birational map surjective in codimension 1, where X, Y are normal $\mathbb{Q}$ -factorial varieties over S . Take a common log resolution $p : W \rightarrow X, q : W \rightarrow Y$. Suppose that $p^*(K_X + B) - q^*(K_Y + \alpha_* B)$ is effective.

1. Prove that if (X, B) is KLT/LC, then $(Y, \alpha_* B)$ is KLT/LC.
2. Give an example that (X, B) is DLT but $(Y, \alpha_* B)$ is not DLT.

Exercise 55. Let $(X, B = \sum_i b_i B_i)$ be a dlt pair, and S an irreducible component of $\lfloor B \rfloor$. Let $K_S + B_S = (K_X + B)|_S$, and $\mathcal{S} = \{1 - \frac{1}{m} + \sum \frac{r_i b_i}{m} \mid m \in \mathbb{Z}_{>0}, r_i \in \mathbb{Z}_{\geq 0}\}$. Show that

$$d(\mathcal{S}) := \sum_{\alpha \in \mathcal{S}} |\{E \mid a(E, S, B_S) < 1 - \alpha, \text{ and } \text{Center}_S(E) \not\subset \lfloor B_S \rfloor\}| < +\infty,$$

where $|\{\cdot\}|$ means the cardinality of the set.

Exercise 56. Consider a $(K_X + B)$ -flip,

$$\begin{array}{ccc} X & \dashrightarrow & X' \\ & \searrow & \swarrow \\ & Y & \end{array}$$

where X, X', Y are normal varieties $f : X \rightarrow Y, f' : X' \rightarrow Y$ are proper and birational. Show that:

1. for any exceptional divisor E over Y , we have

$$a(E, X, B) \leq a(E, X', B'),$$

and the inequality is strict if and only if $\text{Center}_X(E) \subseteq \text{Exc}(f)$, and

2. $\dim \text{Exc}(f) + \dim \text{Exc}(f') \geq \dim X - 1$.

Exercise 57. Show that any sequence of 3-folds flips for terminal pair (X, B) terminates.

Exercise 58. Consider a commutative diagram

$$\begin{array}{ccc} X & \dashrightarrow & X' \\ & \searrow & \swarrow \\ & Y & \end{array}$$

where X, X', Y are normal varieties and $f : X \rightarrow Y, f' : X' \rightarrow Y$ are proper and birational. Let B (resp. B') be an $\mathbb{R}$ -divisor on X (resp. X'). Assume that

1. $f_*B = f'_*B'$,
2. $-(K_X + B)$ is $\mathbb{R}$ -Cartier and f -nef, and
3. $K_{X'} + B'$ is $\mathbb{R}$ -Cartier and f' -nef.

Then for an arbitrary exceptional divisor E over Y , we have

$$a(E, X, B) \leq a(E, X', B')$$

Strict inequality holds if either

- $-(K_X + \Delta)$ is f -ample and f is not an isomorphism above the generic point of center E , or
- $(K_{X'} + B')$ is f' -ample and f' is not an isomorphism above the generic point of center E .

Exercise 59. Let $(X, B) \rightarrow S, (X, B') \rightarrow S$ be two pairs satisfying the BCHM condition. Suppose that $K_X + B'$ is nef over S . Suppose that we can run $(K_X + B)$ -MMP over S with scaling of $B' - B$ (that is, flip exists), and denote $\alpha_i : X \dashrightarrow X_i$ to be the natural birational morphism to the i -step of the MMP. Show that for any $i \neq j$, $\alpha_i \neq \alpha_j$, that is, the induced birational map $\alpha_j \circ \alpha_i^{-1} : X_i \rightarrow X_j$ is not an isomorphism.

Exercise 60. Let $(X, B) \rightarrow S$ be a pair satisfying the BCHM condition. Suppose that it has a minimal model. Show that it has a canonical model. Show that $K_X + B \sim_{\mathbb{R}, S} D \geq 0$.

Exercise 61. Let $f : X \dashrightarrow Y$ be a birational map between two projective normal varieties that is isomorphic in codimension 1. Let D and D_Y be nef $\mathbb{R}$ -Cartier divisors on X and Y respectively. Suppose that D_Y is the strict transform of D on Y . Let $p : W \rightarrow X, q : W \rightarrow Y$ be a common resolution. Show that $p^*D = q^*D_Y$.

Exercise 62. Let $(X, B), (X, B')$ be two $\mathbb{Q}$ -factorial dlt pairs. Suppose that there exists a positive real number λ , such that $K_X + B \equiv \lambda(K_X + B')$.

1. If both (X, B) and (X, B') are klt pairs, then show that (X, B) and (X, B') share the same minimal models.
2. If $\lfloor B \rfloor = \lfloor B' \rfloor$, then show that (X, B) and (X, B') share the same minimal models.

Exercise 63. Let $f : X \rightarrow Y$ be a projective birational morphism.

1. Show that there is an effective $\mathbb{Q}$ -divisor F such that $-F$ is f -ample.
2. Assume that Y is $\mathbb{Q}$ -factorial. Show that there is an effective f -exceptional divisor F such that $-F$ is f -ample.